

Jianing Zhang

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EDUCATION

Carnegie Mellon University

Master of Science in Computer Vision, Robotics Institute

Pittsburgh, PA

December 2027

University of Washington

Bachelor of Science in Computer Science, Cum Laude, Honors in Computer Science | GPA: 3.94/4.00

Seattle, WA

June 2026

RESEARCH INTERESTS

Multimodal Language Models: Integrating vision, text, and other modalities for grounded understanding and reasoning

3D/4D Understanding: Representation learning and generative modeling for spatiotemporal and 3D/4D data

Embodied AI: Agents that perceive, reason, and act in interactive and simulated environments

PUBLICATIONS

Jianing Zhang*, Chenhao Zheng*, Yajun Yang, Max Argus, Rustin Soraki, Winson Han, Taira Anderson, Chun-Liang Li, Shuo Liu, Jiafei Duan, Zhongzheng Ren, Jieyu Zhang, Ranjay Krishna. "MolmoMotion: Forecasting Point Trajectories in 3D with Language Instruction." **NeurIPS 2026 (Spotlight)** (*equal contribution)

Chenhao Zheng, Jieyu Zhang, **Jianing Zhang**, Weikai Huang, Ashutosh Kumar, Quan Kong, Oncel Tuzel, Chun-Liang Li, Ranjay Krishna. "TrajTok: Learning Trajectory Tokens Enhances Video Understanding." **CVPR 2026 (Poster)**

RESEARCH EXPERIENCE

Carnegie Mellon University, Robotics Institute, Katerina Fragkiadaki's Lab

Pittsburgh, PA

Graduate Research Assistant

July 2026 - Present

- Building a unified 3D point world model to learn physics dynamics of rigid and deformable objects by training particle-based dynamics networks on simulated interactions in PyTorch
- Targeting generalization to unseen textures and material properties by curating training data across material types and conditioning dynamics on material cues

Allen Institute for Artificial Intelligence (AI2)

Seattle, WA

Student Researcher

January 2026 - July 2026

- Co-led the MolmoMotion effort, a language-guided 3D motion forecasting model with autoregressive and flow-matching heads in PyTorch, beating every baseline on PointMotionBench (**NeurIPS 2026 Spotlight, co-first author**)
- Created a record-scale action-described 3D point trajectory dataset (1.16M videos, 736 motion types) by building a scalable annotation pipeline with 3D point tracking and language labeling
- Raised robot manipulation success by 20.3% over Molmo2 and won 4 of 5 video quality metrics against Wan2.2-I2V-A14B by transferring forecasted motion into downstream policies and video generators

University of Washington, Paul G. Allen School of Computer Science and Engineering, RAIN Lab

Seattle, WA

Student Researcher Mentors: Chenhao Zheng, Ranjay Krishna

April 2025 - July 2026

- Strengthened video understanding by 4.8% on Kinetics-400 and 4.1% on SSv2 by contributing to TrajTok, a tokenizer replacing patch tokens with object-centric trajectory representations (**CVPR 2026 Poster**)
- Boosted probing accuracy to 82.5% on Kinetics-400, ahead of linear, attentive, and Perceiver baselines, by engineering TrajAdapter components for efficient trajectory feature use
- Lifted long-horizon video reasoning by 8.8% on LongVideoBench by integrating TrajTok into video-language models (TrajVLM) within a fixed token budget

Shanghai Artificial Intelligence Laboratory and Shanghai Jiao Tong University

Shanghai, China

Student Researcher Mentor: Fang Yan; Supervisor: Wei Liu

June 2023 - September 2023

- Led a pathology AI study on whole-slide tumor and cancer subtype classification, benchmarking 3 deep learning architectures to guide clinical model selection
- Surpassed an 80% doctor baseline with 91% tumor detection accuracy on private lung cancer data by fine-tuning attention-based networks first trained on public TCGA kidney data

- Reached 98% AUC on tumor subtype detection by training dedicated classifiers, strengthening diagnostic support in clinical workflows

PROJECTS

Jane Street Market Data Forecasting Competition | Kaggle

November 2024 - January 2025

- Led a team to finish within 0.003 of the winning score by training Transformer and XGBoost forecasters on 40M+ rows of market time-series data with extensive hyperparameter tuning
- Cut memory and GPU load to fit strict hardware limits by streaming data in chunks and optimizing training loops in Python

CIRCLE Official Application | University of Washington

June 2023 - June 2024

- Spearheaded development of UW CIRCLE's official cross-platform app, shipping on Android and iOS, by building it in React Native, React, and JavaScript
- Shaped features around interviews and surveys with 200+ students, improving communication between international students and university departments and reducing misinformation

SKILLS

Programming Languages: Python, C++, C, Java, TypeScript, Shell

ML Frameworks and Libraries: PyTorch, TensorFlow, NumPy, Hugging Face (Transformers, Diffusers, Accelerate), PyTorch3D, OpenCV, scikit-learn

Machine Learning: Diffusion Models, Flow Matching, Transformers, Attention Mechanisms, Point Cloud Learning, Learned Physics Simulation, Reinforcement Learning

Training and Optimization: Multi-GPU Distributed Training (DDP), Mixed Precision, torch.compile, Gradient Checkpointing

Tools and Platforms: Git, Docker, Linux, Slurm, HPC clusters, Weights & Biases, MuJoCo, AWS, GCP, React, React Native, Node.js

Languages: English (Fluent), Mandarin Chinese (Native)